

B-1-A

Roll No.....

Total No. of Questions : 21]

[Total No. of Printed Pages : 4

XIISZRJDF20
1101-A
PHYSICS

Time : 3 Hours]

[Maximum Marks : 70

(Long Answer Type Questions)

5 each

1. Using Gauss's law derive an expression for electric field due to uniformly charged thin spherical shell at a point outside the shell.

Or

✓ What is parallel plate capacitor ? Derive an expression for the energy stored in a capacitor.

- ✓ 2. Give the principle, construction and working of a moving coil galvanometer.

Or

Discuss the properties of dia, para and ferromagnetic materials.

- ✓ 3. What is meant by total internal reflection ? State its conditions.

Or

Define fringe width. Derive expression for fringe width in interference pattern.

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Turn Over

B-1-A

4. Define Rectification. How can a p-n junction diode be used as a full wave rectifier ?

Or

- ✓ Draw an OR gate using two diodes, explain its working and write its truth table.

(Short Answer Type Questions)

3 each

- ✓ 5. Calculate the potential energy in case of dipole having magnitude of each charge as 3×10^{-6} C. The charges are separated at a distance of 2000 Å.
- ✓ 6. Distinguish between resistance and resistivity.
- ✓ 7. Explain the limitations of cyclotron.
- ✓ 8. State and explain Faraday's laws of electromagnetic induction.
- ✓ 9. Write any six characteristics of electromagnetic waves.
- ✓ 10. For a given medium, the polarising angle is 45° . What will be the critical angle of the medium ?
- ✓ 11. State postulates of Bohr's theory of Hydrogen atom.
- ✓ 12. What is space wave propagation ? Give two examples of communication system which use space wave mode.

(Very Short Answer Type Questions)

2 each

- ✓ 13. Briefly explain the principle of Potentiometer.
- ✓ 14. Why is the core of a transformer laminated ?
- ✓ 15. A capacitor behaves as perfect conductor for high frequency a.c. Explain why ?

16. Find the power of a convex lens of focal length 20 cm.
17. Why does sky look blue ?
18. Give *two* points of difference between nuclear fission and nuclear fusion.
19. State laws of photoelectric emission.
20. What is Modulation ?

(Objective Type Questions)

1 each

21. (a) Do as directed :

(i) The copper strips in a slide wire bridge are thick because
..... (Fill in the blank)

(ii) The advantage of placing the prism in *minimum deviation* position is to obtain pure spectrum. (True/False)

(iii) The momentum of a photon is 'P', the wave length is
..... (Fill in the blank)

(iv) Mass defect is difference between
(Fill in the blank)

(v) The resistance and depletion layer in p-n junction diode decreases during forward bias. (True/False)

(b) Choose the correct/most appropriate answer :

(vi) 1 kWh is equal to :

- | | |
|---------------------------------|------------------------------------|
| (A) $3.6 \times 10^5 \text{ J}$ | (B) $3.6 \times 10^{-6} \text{ J}$ |
| (C) $3.6 \times 10^6 \text{ J}$ | (D) 36000 J |

(vii) Resonant frequency of LCR-resonant circuit is :

(A) $2\pi\sqrt{LC}$

(B) $2\pi/\sqrt{LC}$

(C) $\frac{1}{2\pi}\sqrt{LC}$

☒ (D) $\frac{1}{2\pi\sqrt{LC}}$

(viii) The tip of a needle does not give a sharp image on a screen. This is due to :

(A) Polarization

(B) Interference

☒ (C) Diffraction

(D) Refraction

(ix) An electron of mass ' m ' and charge ' e ' is moving from rest through a potential difference ' V ' in vacuum. Its final velocity is :

☒ (A) $\sqrt{\frac{2eV}{m}}$

(B) $\sqrt{\frac{eV}{m}}$

(C) $\frac{eV}{2m}$

(D) $\frac{eV}{m}$

(x) A hole in a p -type semiconductor is :

(A) an excess electron

☒ (B) a missing electron

(C) a missing atom

(D) a positive ion